LightStress: Targeting Stress Reduction through Affective Objects

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Introduction

- LightStress is a **tangible affective artifact**, which can **help reduce stress** levels.
- Long-term **stress** is a major contributor to global **health loss**.
- It has a clear **influence on social**, intellectual, and **economic development**.
- Psychosocial stress is related to **job insecurity** and the quality of work environments.
- **Mental health** continues to be **neglected** in **business settings**.

Aims

- **Mitigate stress** and help **improve mental well-being** in shared workspaces.
- Design a **subtle tangible affective artifact** that **blends in** in the environment in which it is used.

Methods

- A **questionnaire**, to collect relaxing common elements among **11 participants**, 8 females and 3 males, aged 22 to 45 (M = 28.8, SD = 6.6).
- The questionnaire focused on how **colors, sounds and places** impact calm and relaxed states of mind.
- **Imagery research** to analyze **types of objects used to personalize** people's **professional environments**, which informed the design of our prototype.

Results

- Most **individuals take refuge in their home when feeling stressed** and associate **nature--related sounds** to a relaxed state of mind.
- 9 users indicated **blue** as a calming color, 6 **green**, and 5 **yellow**.
- **3 object typologies** were found: paper-based products, decorative, and **miniatures**, such as figurines and other memorabilia.

ACKNOWLEDGMENTS
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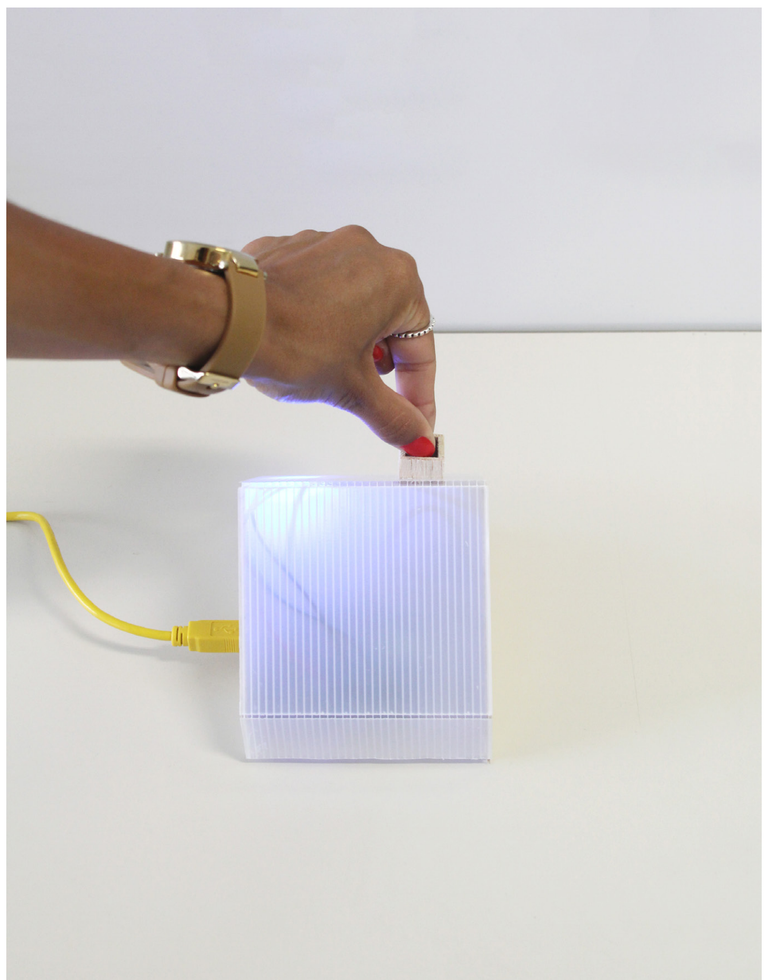

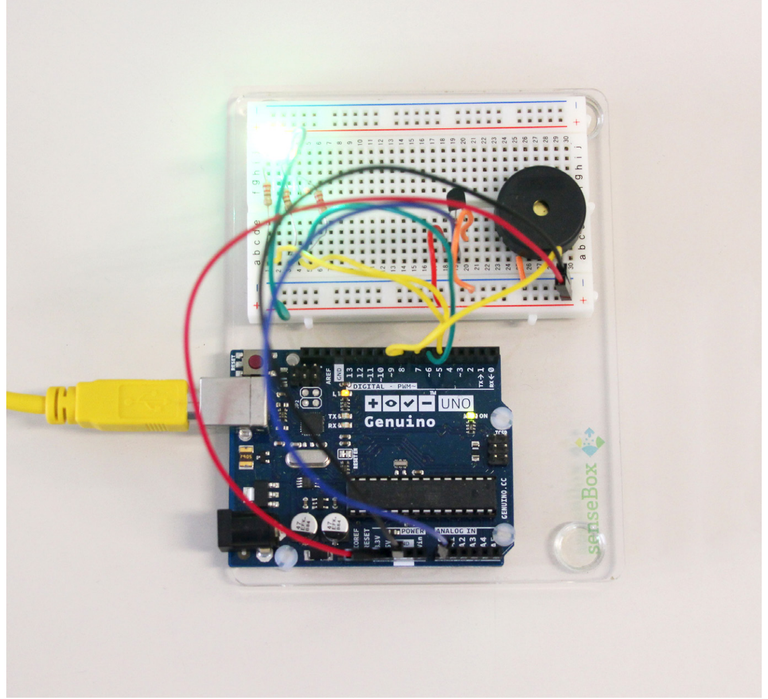

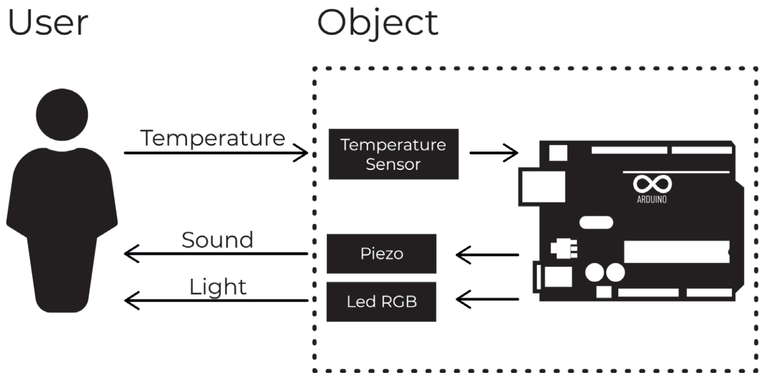

Stress Level	Temperature	Color output	Sound output
High	< = 21 °C	Blue	Water
Moderate	22° to 25 °C	Green	None
Normal	> = 25 °C	Yellow	None

Conclusions

- The LightStress prototype can **help reduce stress** by combining color changing LED to different sounds, controlled by the user's body temperature.
- Our approach **goes against the mainstream methods** for measuring and managing stress, as it **draws insight from pre-existing coping mechanisms** to deliver an **unobtrusive system** that can be used at the workplace.

Next steps

- Explore additional affective interactions.
- Exchange the Piezo Buzzer for a miniature portable MP3 player (allowing more meaningful audio feedback).
- Replace tones for audio clips with water sounds (i.e. ocean and rain), which have a soothing effect during stressful situations (as reported by 7 out of 11 participants).

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